

EmberGym: A Geographically Grounded Benchmark for Hive-Mind Wildfire Suppression with Reinforcement Learning

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Abstract. Effective wildfire suppression is essential for enabling rapid response and limiting the spread of destructive fire events. Reinforcement learning, though a potential methodology for autonomous wildfire suppression, lacks a common benchmark. Prior studies train on synthetic grids disconnected from real geography, compare mainly against uncontrolled baselines, and ignore deployment constraints. This paper presents EmberGym, a framework that builds suppression environments from real geospatial data with a vectorised dual-fidelity fire engine. A per-cell spatial action interface, live-weather inference, digital-twin replay, and a quantization toolkit is used. The work introduces a HiveMind system in which drones and tankers are driven by a single learned U-Net intent map coupled with combinatorial entity-to-target assignment. Tests performed on over 60 seeded episodes of HiveMind significantly outperforms a matched hand-coded hive (paired Wilcoxon $p < 10^{-10}$), winning 58 of 60 seeds with a small fleet and 60 of 60 with a larger fleet, where it preserves $2.5\times$ more fuel. Ablations show the gain does not come from forecast channels (a null result) and that greedy dispersed assignment slightly beats Hungarian matching. Policies transfer across fidelity tiers, and static INT8 quantization makes the hive brain $3.7\times$ smaller and $1.6\times$ faster with no accuracy loss.

Keywords: Wildfire suppression · Reinforcement learning · Multi-agent systems · Benchmarks · Quantization

1 Introduction

Wildfires are a persistent global crisis; between 2000 and 2020 nearly 1.2 billion people lived in regions of immediate wildfire risk [1]. Autonomous suppression is a sequential decision problem to which deep RL is increasingly applied [2–4], yet four gaps limit progress. *Geographic realism:* most Reinforcement learning (RL) studies use synthetic grids disconnected from real terrain and fuels. *Incompatible environments:* each paper ships its own simulator, and existing benchmarks lack realism and difficulty [8]. *Weak baselines:* agents are usually compared against

uncontrolled spread, whereas informed heuristics, and at scale mixed-integer optimisation [8,9], set a far higher bar. *Deployment blindness*: policies are evaluated in FP32, ignoring edge latency, memory, and energy limits.

This research presents EmberGym, which builds environments from LANDFIRE data [11]. The system steps fire with a vectorised engine offering matched cellular-automaton and Rothermel-based [12] fidelity tiers. This proposed system also fixes an evaluation protocol (seeds 900 to 959, fixed metrics, heuristic baselines). This research proposes HiveMind: a framework of heterogeneous entities driven by one shared U-Net [13] emitting a global *intent map*. A combinatorial step dispatches each entity based on this map. Study uses the predict-with-learning, prescribe-with-optimisation recipe of operational practice [9] and system is made end-to-end trainable. This research contributes to the framework and protocol, the HiveMind architecture, a matched forecast ablation, a Hungarian versus greedy assignment study, fidelity-transfer and architecture-dependent quantization results.

There has been multiple RL solutions in the past for wildfire suppression. Deep multi-agent RL have scaled where exact solutions have not been able to deliver [2]. DQN variants beat heuristics by only 1.6 to 1.7 percent on firebreak placement [3]. Rainbow DQN [5] has driven drone swarms [4], and SimHarness [6] and FireCastRL [7] add Rothermel spread and real terrain to the model. But, none of these solutions provide a shared protocol with per-seed statistics. In optimisation, suppression allocation is NP-complete and reaches state of the art with mixed-integer programming [8] and predict-then-prescribe frameworks [9]. HiveMind integrates these optimization principles into a RL paradigm, enabling end-to-end learning of suppression policies. Finally, QuaRL [10] quantized RL policies to 8 bits without reward loss. This work showcases the first application to wildfire suppression and demonstrates that it is architecture dependent.

2 Methodology

The proposed EmberGym Framework has five layers. **Data:** a loader ingests LANDFIRE GeoTIFFs into per-class ignition, spread, and burn parameters, with a procedural generator producing statistically matched terrain for download-free reproduction. Weather providers cover stochastic training weather, recorded digital-twin replay (scored by intersection over union against observed perimeters), and a live adapter. **Simulation:** This work implements a vectorised eight-neighbour engine (microsecond timesteps and GPU portable). It has two tiers on one code path: *fast mode* based on a cellular automaton modulated by wind, slope, humidity, temperature, moisture; and *standard mode* with a Rothermel-style rate-of-spread mapping. **Environments:** Implements a Gymnasium environment with a nine-channel observation. Novel to EmberGym is a *spatial action mode* (a per-cell score map: the environment executes the top- k legal cells with masking), plus a PettingZoo-style multi-agent wrapper. **Agents:** no-agent, random, and greedy-heuristic baselines are used. A compact Rainbow (double,

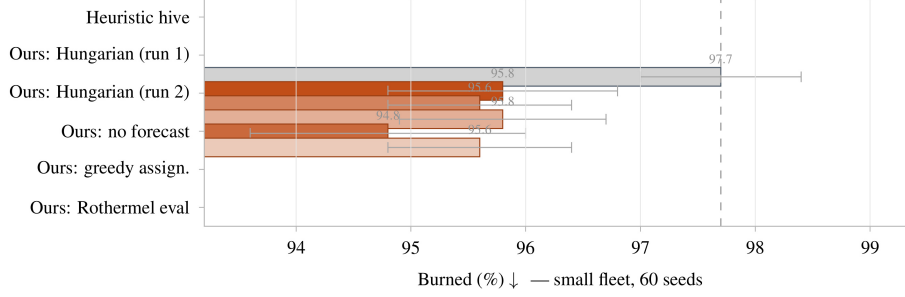


Fig. 1: HiveMind variants (mean $\pm$ s.d., axis truncated). The two Hungarian runs bound training-seed variance at ~ 0.2 pts, so the forecast ablation is a null result; greedy dispersed assignment is slightly *better* than Hungarian matching ($p < 0.001$); and zero-shot evaluation on the Rothermel tier is indistinguishable from in-tier performance. Dashed line: matched heuristic hive.

dueling, noisy [14], n -step, prioritised replay), and the U-Net policy, with INT8 quantization and ActorQ-style rollouts [10] are used for deployment.

HiveMind: Entities are physical units without policies: drones (speed 4, payload 3, single-cell drop, 6-step refill) and tankers (speed 1, payload 12, radius-2 splash, 10-step refill). A shared U-Net reads the full observation and emits one intent map $I \in \mathbb{R}^{H \times W}$. Entities are matched to distinct targets by maximising:

$$\sum_i [I(c_i) - \lambda \tau(e_i, c_i)], \quad \tau(e, c) = \frac{\|p_e - c\|_\infty}{v_e}, \quad \lambda = 0.15, \quad (1)$$

with τ the entity travel time. The default solver is the Hungarian algorithm [15]; a greedy variant reserves a 3×3 neighbourhood around each target. Coordination is therefore *structural*: unlike MARL approaches in the style of MAPPO or QMIX, no communication protocol must be learned. The actor is trained with advantage actor-critic (a spatial softmax over the intent map, one sampled cell per ready entity, a CNN critic baseline, and entropy regularisation).

Forecast channels: a rolling-window polynomial regression predicts conditions $h \in \{5, 10\}$ steps ahead, spatialised via the lapse rate (0.65°C per 100 m) into per-cell ignition-risk channels. Matched heuristic and forecast-heuristic hives drive the identical fleet through the identical assignment machinery, so agent differences are purely cognitive.

3 Experiments and Results

All runs use a 64×64 synthetic Sierra-Nevada-like terrain (seed 42), three random ignitions per episode, the fast tier for training, 100-step episodes, and greedy evaluation over the 60 fixed seeds 900 to 959 shared across agents for paired statistics. HiveMind trains for 2000 episodes per configuration (Adam, learning rate of 3×10^{-4}) with small (2 drones, 1 tanker) and large (4 drones, 2 tankers) fleets. The reward penalises each newly burned cell (-0.3 , with -2.5 for settlement cells) with terminal bonuses, and the severe (hot, dry, windy) regime produces near-total burns without intervention.

Table 1: Main benchmark on the 64×64 grid over 60 seeded episodes (mean $\pm$ standard deviation). Burned percentage is the primary metric (lower is better); Front. denotes frontline efficiency; Suppr. denotes suppression efficiency versus the no-agent baseline

Agent	<i>Small</i> (2 drones, 1 tanker)			<i>Large</i> (4 drones, 2 tankers)		
	Burned	Front.	Suppr.	Burned	Front.	Suppr.
No agent	99.7 ± 0.1	—	0.0	99.7 ± 0.1	—	0.0
Heuristic hive	97.7 ± 0.7	92.0	2.0	96.3 ± 1.1	88.3	3.4
Forecast-heur. hive	97.8 ± 0.8	83.7	1.8	96.2 ± 1.2	81.2	3.5
HiveMind (ours)	95.8 ± 1.0	25.5	3.9	90.6 ± 2.0	22.0	9.1

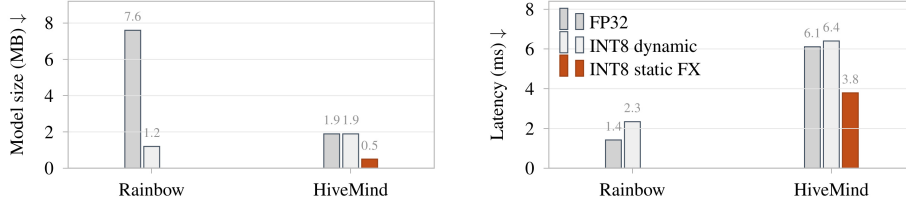


Fig. 2: Quantization is architecture dependent. Dynamic PTQ compresses the Linear-heavy Rainbow net $6.3\times$ but *slows* it; on the fully convolutional hive brain it is a silent no-op (identical size). Static graph-mode INT8 gives $3.7\times$ compression and $1.6\times$ speed-up at 95.4 ± 0.9 vs. $95.7 \pm 0.9\%$ burned.

Results show mean $\pm$ standard deviation and paired two-sided Wilcoxon signed-rank tests. Each configuration was trained twice to estimate training-seed variance.

Results. Table 1 shows results with the small fleet HiveMind burns 95.8 ± 1.0 percent versus 97.7 ± 0.7 percent for the matched heuristic hive (median -2.0 points, $p = 2.6 \times 10^{-11}$), winning on 58 of 60 seeds; with the large fleet the gap widens to 90.6 ± 2.0 versus 96.3 ± 1.1 percent (median -5.5 points, $p = 1.6 \times 10^{-11}$), winning on 60 of 60 and preserving $2.5\times$ more intact fuel (9.4 versus 3.7 percent). Suppression efficiency is 9.1 versus 3.4 percent: the learned hive gains from added entities while the heuristic saturates. Notably HiveMind wins with *low* frontline efficiency (22 to 26 percent versus 88 to 92 percent), placing water pre-emptively rather than chasing the front. Settlement survival stays poor throughout (at most 2.2 percent), which the heuristic’s proximity bonus protects slightly better, a reward-shaping limitation.

Ablations. Removing forecast channels leaves HiveMind unchanged (95.8 ± 0.9 percent versus 95.8 ± 1.0 and 95.6 ± 0.8 for two forecast-equipped runs, within training-seed variance; the zero-training version agrees, 97.8 versus 97.7 percent). A null result attributed to near-stationary synthetic weather. Contrary to expectation, greedy dispersed assignment slightly beats Hungarian (94.8 ± 1.2 versus 95.8 ± 1.0 percent, $p < 0.001$). The greedy 3×3 reservation enforces dispersion, whereas Hungarian optimises the myopic per-step objective of Eq. (1) and may cluster entities. Optimal assignment of an imperfect instantaneous objective need not be episode-optimal.

Transfer and deployment. Evaluated without retraining on the Rothermel tier, HiveMind burns 95.6 ± 0.8 percent (indistinguishable from in-tier), so the learned intent generalises across spread models. Dynamic INT8 compresses the Linear-heavy Rainbow

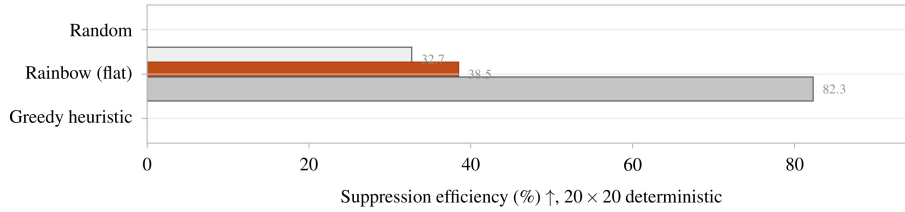


Fig. 3: Single-agent reference: an informed heuristic remains more than twice as strong as flat-action value-based RL, motivating the spatial action interface and the hive architecture.

Q-network $6.3\times$ (7.6 to 1.2 MB) but raises latency on the small 20×20 network (1.42 to 2.34 ms). This is a silent no-op on the fully convolutional U-Net, since it targets only linear layers. Static graph-mode INT8 with activation calibration resolves this: the actor shrinks $3.7\times$ (1.89 to 0.50 MB) and accelerates $1.6\times$ (6.11 to 3.79 ms) at 95.4 ± 0.9 versus 95.7 ± 0.9 percent for FP32. There is no degradation, so the correct mode is architecture dependent.

Limitations. Training curves plateau early and frontline efficiency is low, so study cannot fully separate HiveMind’s advantage into learned intent versus the assignment machinery without an untrained-network control. Each ablation is also a single run (the duplicated Hungarian configuration bounds variance at about 0.2 points). The severe regime compresses all agents into a high-burn band. The Rothermel tier omits fire-atmosphere coupling, and the synthetic terrain is not real terrain.

4 Conclusion

EmberGym provides what RL wildfire research has lacked: a geographically grounded, seeded, heuristic-baselines benchmark with deployment tooling. HiveMind shows that centralised learned valuation with structural coordination significantly outperforms a matched reactive heuristic by a margin that grows with fleet size. While the ablations honestly report that forecasts contributed nothing and that optimal myopic assignment is not the right target.

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